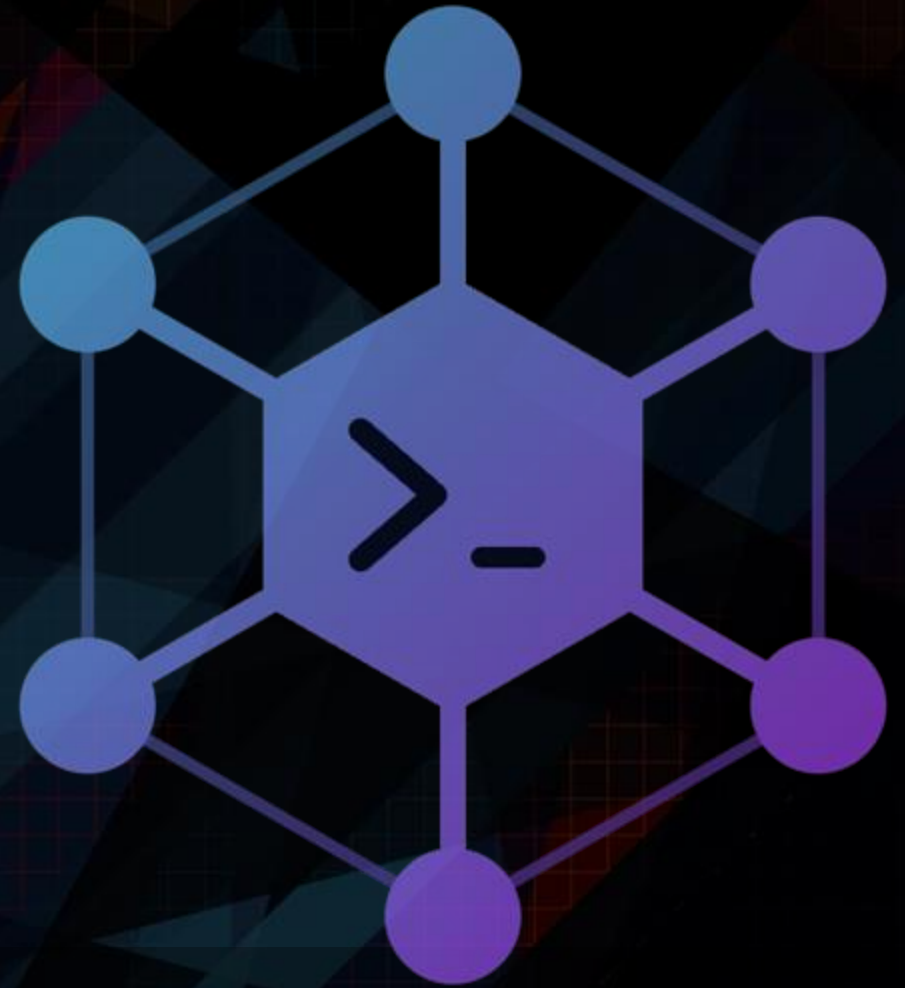


AI Task Agent Using More Flexible Generic Tools via Model Context Protocol (MCP)

Milestone 1

*Jadon Duff
Jacob Ewasko
Ioannis Limniatis
Jacob Lebkuecher*



Milestone 1 Progress Report

Task	Completion %	Jadon D	Jacob E	Jacob L	Ioannis L	To Do
1. Compare and Select Technical Tools	100%	25%	25%	25%	25%	None
2. “Hello World” Demos	100%	25%	25%	25%	25%	None
3. Resolve Technical Challenges	100%	25%	25%	25%	25%	None
4. Compare and Select Collaboration Tools	100%	25%	25%	25%	25%	None
5. Requirement Document	100%	50%	20%	10%	20%	None
6. Design Document	100%	20%	10%	50%	20%	None
7. Test Plan	100%	15%	50%	15%	20%	None



Compare & Select Technical Tools

- *To compare and select technical tools, team members made a matrix for each tool with pros and cons. The team then voted.*
- Linux: Ubuntu Server LTS
- Communications: Transfer Folder
- Fast Context: grep MCP Tool
- Agentic Interaction with Browser: Playwright
- Agentic Interaction with Email: Mail Shadow MCP Server
- Agentic Interaction with GUI: PyAutoGUI
- Agentic GUI Vision: Screenshot, YOLO
- Dynamic Memory: Custom Built
- Base LLM: Qwen3.8-27b
- Deployment Environment: Docker
- Retrieval Augmented Generation: Qwen3-Embedding-8B



Hello World Demos

- Done for all technical tools
- Demoed in-person/virtually
- All tools verified to work
- Ubuntu Server LTS, Transfer Folder, grep MCP Tool, Playwright, Mail Shadow MCP Server, PyAutoGUI, Screenshot, YOLO, Qwen3.8-27b, Docker, Qwen3-Embedding-8B



Hello World Demos

```
(.venv) PS C:\Users\Yianni\Desktop\Project> python .\contextWindow.py
--- History ---
[1] Hello World
[2] How are you?
[3] I am fine.
[4] What is the weather?
[5] It is sunny.
5 messages, ~15 tokens, budget 8 tokens

--- Sliding Window ---
Kept:      ['What is the weather?', 'It is sunny.']
Dropped:   ['Hello World', 'How are you?', 'I am fine.']

--- Compaction ---
Summary: Earlier: Hello World

--- File Memory ---
Saved to memory.json
Recalled: Hello World
(.venv) PS C:\Users\Yianni\Desktop\Project> |
```



Resolve Technical Challenges

- Resolved 23 technical challenges defined in Project Plan
- Difficult challenges were noted down in team Confluence, others discussed with entire team
- Examples:
 - Is the MCP properly sandboxed?
 - To properly sandbox the MCP, we will need to use an established framework and ensure that the tools are mapped properly with no edge cases. Using a common framework like FastMCP will cover this for us.
 - What if the LLM is not good at searching context with the tool?
 - If the LLM is not good at searching context with the tool, we can first try changing the system prompt and hyperparameters. If these are unsuccessful, then we will require a different LLM to be used.



Compare & Select Collaboration Tools

- *To compare and select collaboration tools, team members made a matrix for each tool with pros and cons. The team then voted.*
- Project Management: Atlassian Jira
- Knowledge Management: Atlassian Confluence
- Version Control/Hosting: GitHub
- Task Calendar: Atlassian Jira
- Documentation: Atlassian Confluence
- Presentations: Microsoft PowerPoint
- Communication: Discord



System Requirements

	ID	Count	Meaning
Functional Requirements	FTE	39	Functional Task Execution using Generic MCP
	FAI	9	Functional Improved AI Knowledge
	FDP	20	Functional Versatile Deployment
Interface Requirements	IUI	16	Interface User Interface
	ISI	6	Interface Software Interface
Performance Requirements	PFR	13	Performance
	OPS	28	Task Based Performance (Operations)
	ATA	131	Total



Example Requirements

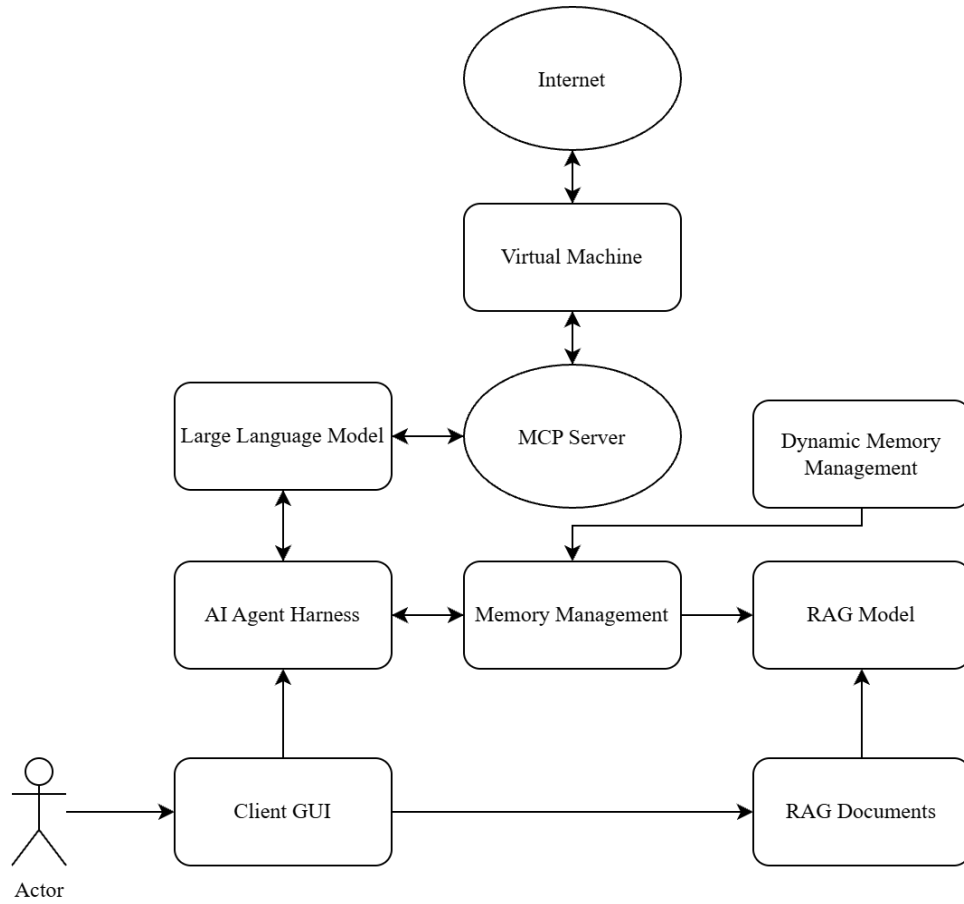
ATA-FTE-011: The system shall allow the AI Agent to monitor running processes on the virtual machine, defined as getting a text output format of the current running processes and ingesting these into the AI Agent's context window.

ATA-IUI-014: The system shall require a single UI interaction to designate an air-gapped system, disabling Internet and other networked/communication abilities.

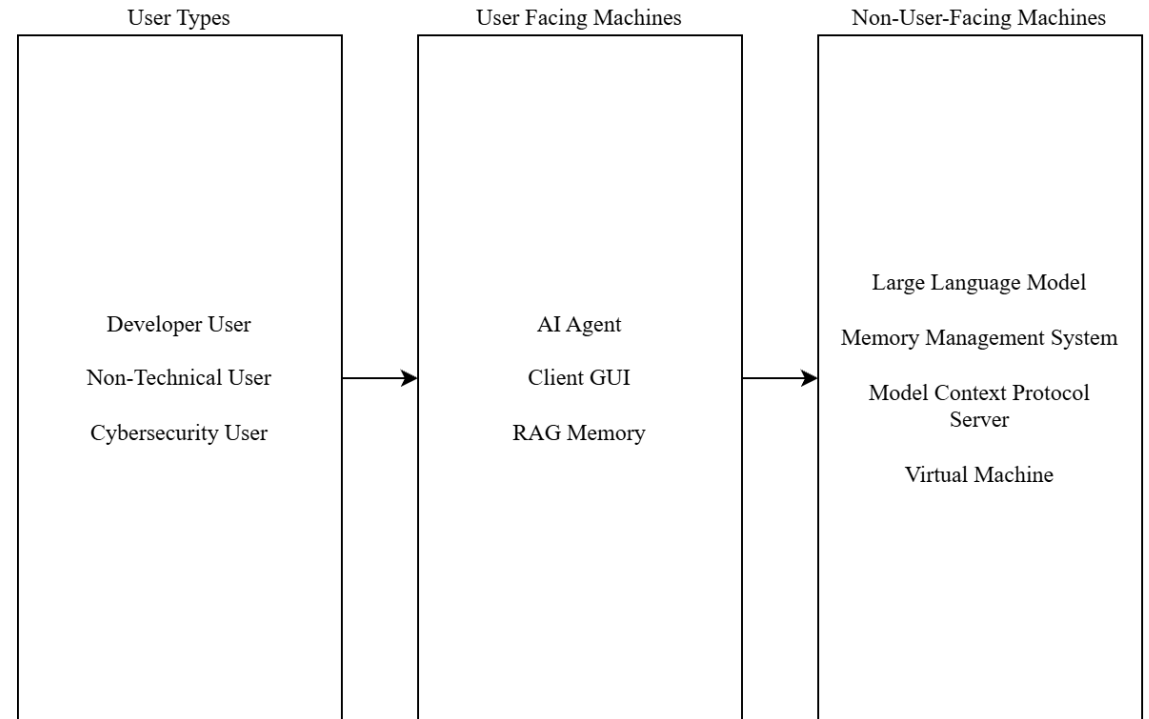
ATA-PFR-010: A User shall submit a Task to the AI Agent within 15 seconds, with the AI Agent starting on the Task within these 15 seconds, not counting the time to type in the prompt.



Design: System Architecture



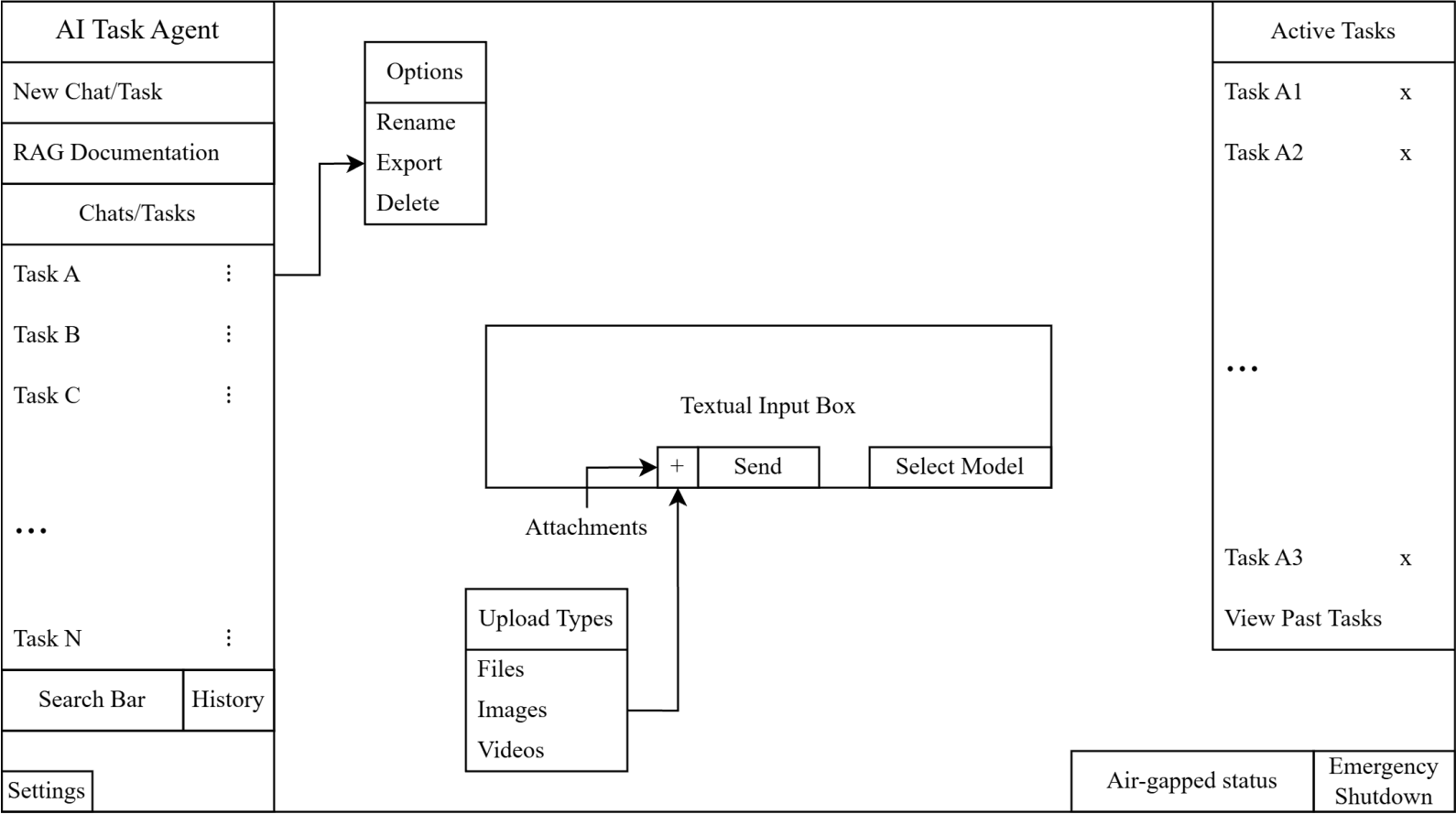
Main Modules



3 Pillars



Design: User Interfaces

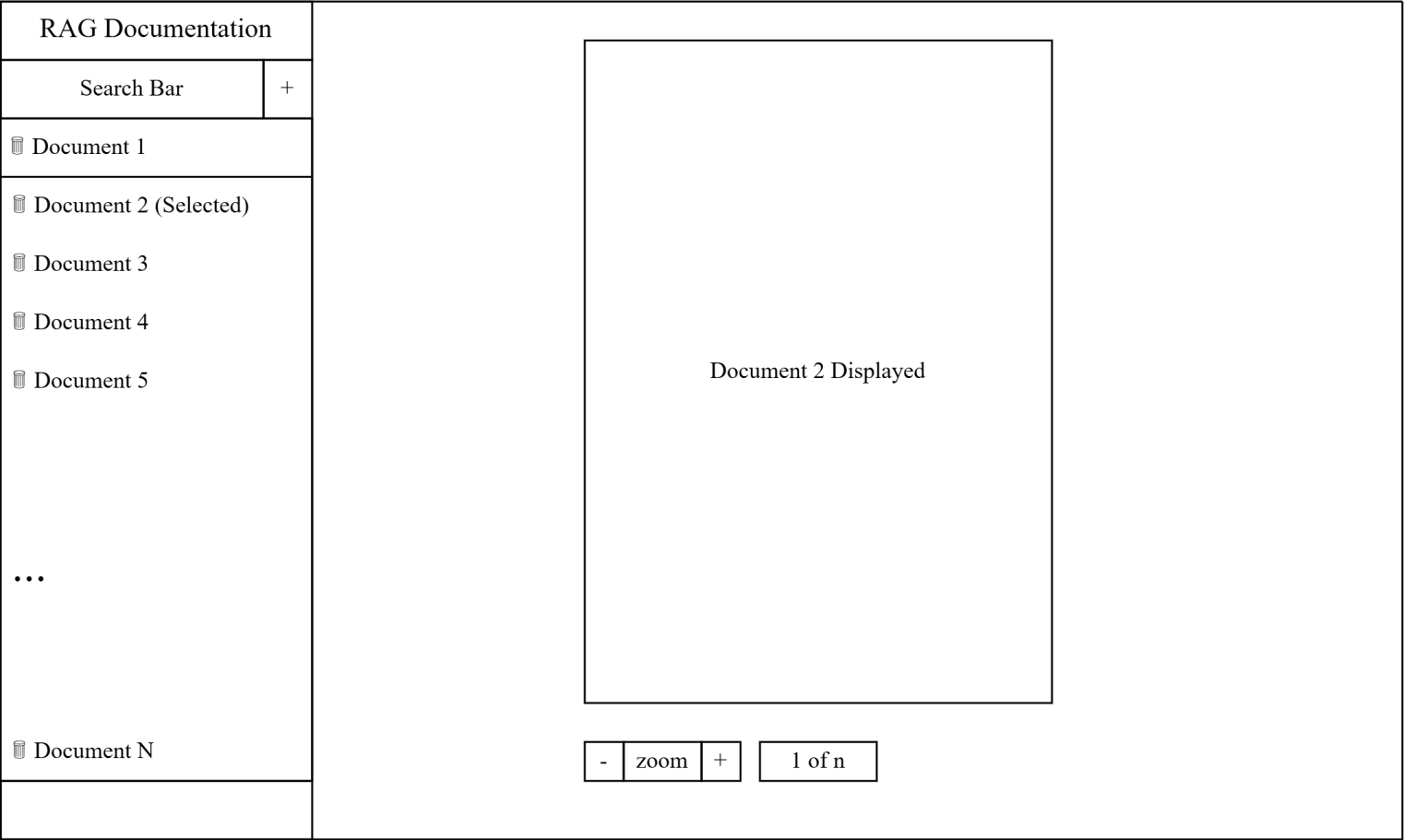


Main Menu

AI Task Agent – Milestone 1



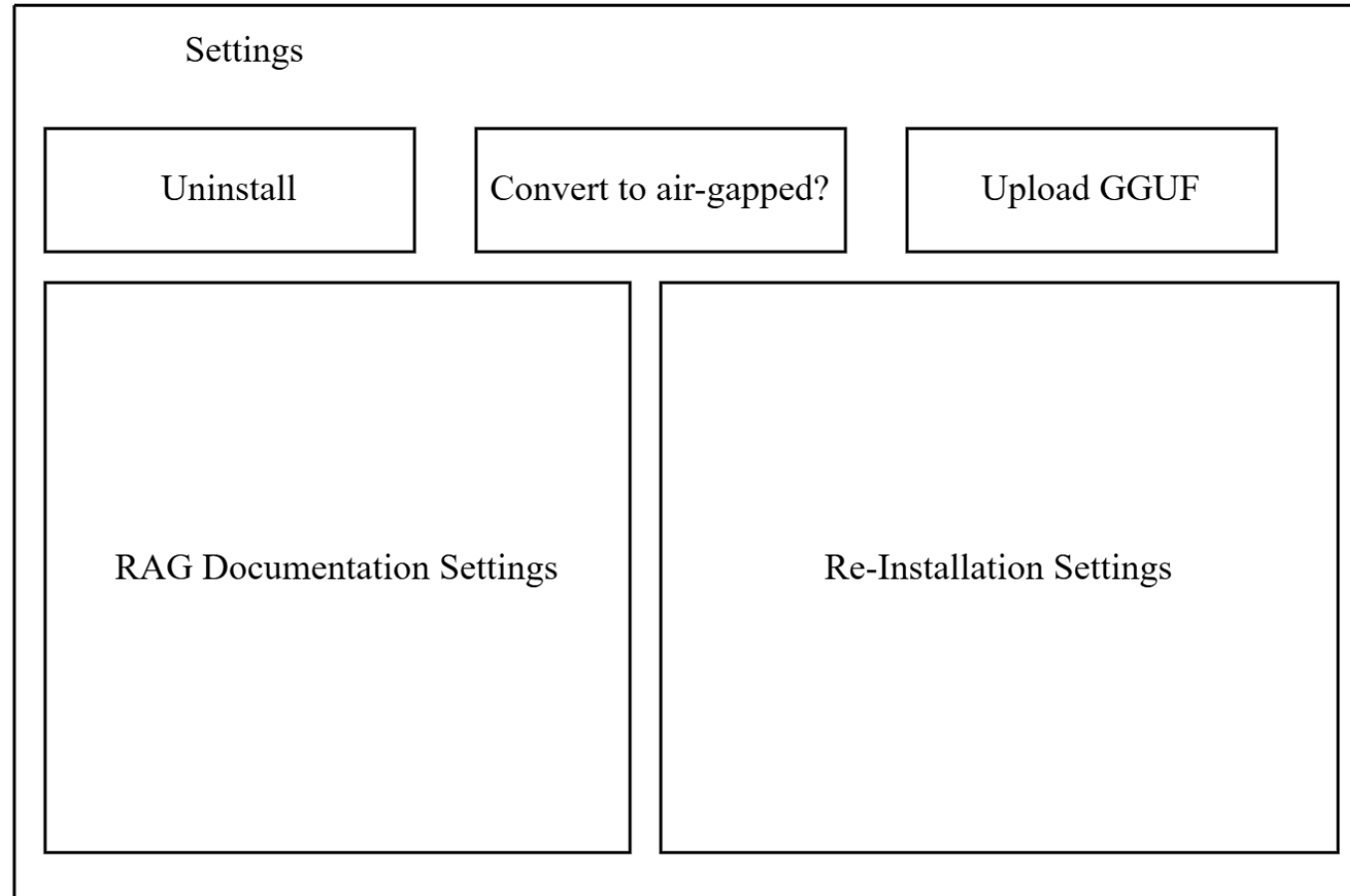
Design: User Interfaces



**RAG Documentation
Menu**



Design: User Interfaces



Settings Menu

AI Task Agent – Milestone 1



Test Plan

	ID	Test Case Count	Meaning
Functional Requirements Tests	FTE	9	Functional Task Execution using Generic MCP
	FAI	5	Functional Improved AI Knowledge
	FDP	7	Functional Versatile Deployment
Interface Requirements Tests	IUI	12	Interface User Interface
	ISI	4	Interface Software Interface
Performance Requirements Tests	PFR	4	Performance
	OPS	15	Task Based Performance (Operations)
	ATA	56	Total



Example Test Cases

Implementation:

- Input:

The user closes AI Task Agent. The user starts up AI Task Agent.

- Desired Output:

AI Task agent closes. AI Task Agent starts up within 120 seconds.

- If this runs correctly, this satisfies:

The software shall start after being closed within a period of 120 seconds, not including the time to load the LLM weights into memory. (ATA-PFR-009)

Implementation:

- Input (Unusual Input):

The user creates a python script called “broken.py” that contains faulty code with the content ‘prin(“hi”)’. Then the user prompts the agent “Run broken.py”

- Desired Output:

Broken.py is run, and an error raises, so the error is queued and routed to AI Task Agent automatically.

- If this runs correctly, this satisfies:

During execution of code, should the program raise an error, this error shall be queued and routed to the AI Agent automatically. Routed to the AI Agent means the error shall be automatically appended to the next prompt to the AI Agent. (ATA-FTE-021)



Milestone 2 Task Matrix

Task	Jadon D	Jacob E	Jacob L	Ioannis L
1. Implement, test, and demo Prototype for Task Execution using Generic MCP Tools, without advanced security features implemented (feature 1).	Implement/test/demo Model Context Protocol tools to meet at minimum 20% of the ATA-FTE requirements.	Implement/test/demo Model Context Protocol tools to meet at minimum 20% of the ATA-FTE requirements.	Implement/test/demo Model Context Protocol tools to meet at minimum 20% of the ATA-FTE requirements.	Implement/test/demo Model Context Protocol tools to meet at minimum 20% of the ATA-FTE requirements.
2. Implement, test, and demo Prototype AI Agent, with full connection to Prototype Task Execution using Generic MCP Tools (feature 1).	Implement/test/demo LLM connected to AI Agent to solve at least 3 of the ATA-OPS requirements.	Implement/test/demo LLM connected to AI Agent to solve at least 3 of the ATA-OPS requirements.	Implement/test/demo LLM connected to AI Agent to solve at least 3 of the ATA-OPS requirements.	Implement/test/demo LLM connected to AI Agent to solve at least 3 of the ATA-OPS requirements.
3. Implement, test, and demo POC RAG memory, with basic documentation tested and linked to Prototype AI Agent (feature 2).	Implement/test/demo Proof-of-Concept RAG system that allows AI Agent to ingest documents and media.	Implement/test/demo Proof-of-Concept RAG system that allows AI Agent to ingest documents and media.	Implement/test/demo Proof-of-Concept RAG system that allows AI Agent to ingest documents and media.	Implement/test/demo Proof-of-Concept RAG system that allows AI Agent to ingest documents and media.
4. Implement, test, and demo Prototype for Versatile Deployment with Widespread Hardware Compatibility, with the system functional to current development on a new install on a new system (feature 3).	Implement/test/demo prototype for deployment. Meaning, implement/test/demo a user interface, allowing users to interact with the system.	Implement/test/demo prototype for deployment. Meaning, implement/test/demo a user interface, allowing users to interact with the system.	Implement/test/demo prototype for deployment. Meaning, implement/test/demo a user interface, allowing users to interact with the system.	Implement/test/demo prototype for deployment. Meaning, implement/test/demo a user interface, allowing users to interact with the system.



Questions

*AI Task Agent –
Milestone 1*

